We are given that for a real quadratic $f : \mathbb{R} \to \mathbb{R}$ and $a,\ b,\ c \in \mathbb{R}$ the following:

$$\begin{cases} f(a) = bc \\ f(b) = ac \\ f(c) = ab \end{cases}$$

Sol. Let $f : x \mapsto kx^2 + qx + p$, for $k,\ q,\ p \in \mathbb{R}$. Hence, we have the following:

$$\begin{aligned} f(a) &= ka^2 + qa + p = bc \\ f(b) &= kb^2 + qb + p = ac \\ f(c) &= kc^2 + qc + p = ab \end{aligned}$$

Consider $f(a) - f(b) = k(a^2 - b^2) + q(a - b)$. Hence:

$$\begin{aligned} f(a) - f(b) &= (a - b)[k(a + b) + q] \\ f(c) - f(a) &= (c - a)[k(a + c) + q] \\ f(b) - f(c) &= (b - c)[k(b + c) + q] \end{aligned}$$

Hence: